

**Notes: Mueller, Osterwalder, Zweimüller, and Kettemann (RES,
2024)**
**Vacancy Durations and Entry Wages: Evidence from Linked
Vacancy–Employer–Employee Data**

Contents

1	Big picture	3
2	Data and measurement	3
2.1	AMS vacancy data and linked worker–firm records	3
2.2	Vacancy duration	3
2.3	Entry wages, wage floors, and AKM components	3
3	Models and empirical specifications	4
3.1	Baseline wage–duration regression	4
3.2	Wage-floor specification	4
3.3	Worker fixed effects	4
3.4	AKM decomposition regression	4
3.5	Contribution of wages to filling-rate variation	4
4	Identification and empirical design	4
5	Tables and figures with detailed interpretation	5
5.1	Figure 1: AMS vacancy data versus representative vacancy survey and Table 1: Vacancy characteristics, AMS universe versus matched sub-sample	5
5.2	Figure 2: Vacancy filling and posting relative to date of availability of job	5
5.3	Figure 3: Log entry wage and log vacancy duration, Table 2: Regressions with log vacancy duration as dependent variable, and Table 3: Worker fixed-effects regressions	6
5.4	Figures 4–6 and Table 4: AKM wage components and vacancy duration	7
5.5	Figure 7: Decision lags versus competition for high-skill workers	9
5.6	Figure 8: Hires-weighted OLS regression of daily filling hazard and starting wages at the establishment growth bin level and Table 5: Simulation results of Kaas and Kircher (2015)	10
6	Short summary	10
7	Conclusion	11
7.1	Core conclusion	11
7.2	What the paper changes conceptually	11
7.3	Implication for search-and-matching models	11
7.4	Interpretation of the small elasticity	11
7.5	Policy and measurement implications	11
7.6	Limitations and open questions	11

7.7 Takeaway	11
------------------------	----

1 Big picture

- **Question.** How are vacancy durations related to the wage that firms pay when they fill the vacancy?
- **Data innovation.** The paper links Austrian public employment service vacancy records to establishment and worker social-security records.
- **Main controlled result.** Once worker/job heterogeneity is controlled for, the relation between vacancy duration and entry wage turns negative. The preferred elasticity is about -0.075 .
- **Wage decomposition result.** The AKM establishment wage component is negatively related to vacancy duration, while worker components are positively related to vacancy duration.
- **Quantitative implication.** Wage policies have the right sign but explain only a small fraction of the variation in vacancy-filling rates across establishments.
- **Economic takeaway.** Hiring standards, screening, recruiting intensity, non-wage amenities, and other non-wage channels likely matter more than wage posting.

2 Data and measurement

2.1 AMS vacancy data and linked worker–firm records

The vacancy data come from the Austrian public employment service (AMS). The data contain vacancy characteristics, posting dates, desired availability dates, filling dates, and establishment identifiers. For mediated vacancies, they can be linked to the worker who filled the vacancy and to Austrian social-security earnings records.

Interpretation. The central contribution is the three-way link: vacancy, posting establishment, and filling worker.

2.2 Vacancy duration

$$d_i = t_i^{filled} - t_i^{available}.$$

The main duration measure is the difference between the filling date and the date when the job becomes available.

Interpretation. This duration concept focuses on the firm’s search problem around the desired start date.

2.3 Entry wages, wage floors, and AKM components

The main wage is the log entry wage. For 2011–2014, the paper also constructs entry wages relative to occupation-specific wage floors. The AKM decomposition is

$$\log w_{jkt} = \alpha_j + \psi_k + x'_{jt}\beta + \varepsilon_{jkt},$$

where α_j is the worker effect, ψ_k is the establishment effect, and ε_{jkt} is the residual.

Interpretation. The AKM decomposition separates high-wage workers from high-wage firms.

3 Models and empirical specifications

3.1 Baseline wage–duration regression

$$\log d_i = \alpha + \beta \log w_i + X_i' \Gamma + \tau_t + \varepsilon_i.$$

Interpretation. β is the elasticity of vacancy duration with respect to entry wages. The sign changes from positive in raw data to negative after controls.

3.2 Wage-floor specification

$$\log d_i = \alpha + \beta \left(\log w_i - \log \underline{w}_{occ(i),t} \right) + X_i' \Gamma + \tau_t + \varepsilon_i.$$

Interpretation. This checks whether the result is driven by collective-bargaining wage floors.

3.3 Worker fixed effects

$$\log d_{it} = \alpha_i + \beta \log w_{it} + X_{it}' \Gamma + \tau_t + \varepsilon_{it}.$$

Interpretation. This compares different unemployment spells for the same worker.

3.4 AKM decomposition regression

$$\log d_i = \alpha + \beta_E \psi_k + \beta_W \alpha_j + \beta_X x'_{jt} \hat{\beta} + \beta_R \varepsilon_{jkt} + X_i' \Gamma + u_i.$$

Interpretation. The establishment wage component is the part most relevant for wage-posting theories.

3.5 Contribution of wages to filling-rate variation

$$\text{contribution} \approx \frac{|\epsilon_{f,w}| \epsilon_{w,h}}{\epsilon_{f,h}}.$$

Interpretation. The paper finds that wages explain only about 1–2.4% of the observed filling-rate/hires-rate relationship.

4 Identification and empirical design

The key empirical challenge is that high wages reflect worker quality, establishment wage premia, and match-specific components. The paper uses occupation fixed effects, worker fixed effects, wage-floor adjustments, AKM decomposition, and linked employer–employee controls to separate these channels. The design is observational, so the estimates are interpreted as conditional associations rather than randomized wage effects.

5 Tables and figures with detailed interpretation

5.1 Figure 1: AMS vacancy data versus representative vacancy survey and Table 1: Vacancy characteristics, AMS universe versus matched sub-sample

Read: Figure 1 shows that AMS vacancy counts closely track the representative OStE vacancy survey, with correlation 0.88. Table 1 compares the AMS universe with the matched worker sample.

Economic message. The AMS data are broadly informative, but the matched wage analysis uses a selective subsample that must be handled carefully.

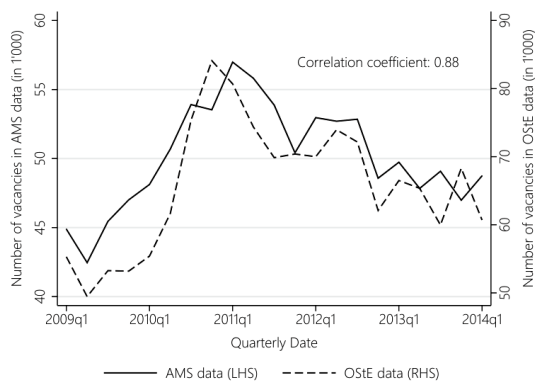


FIGURE 1
AMS vacancy data versus representative vacancy survey
Comparison of the number of AMS vacancies with the number of vacancies in a representative vacancy survey [Offene-Stellung (OStE)] 2009q1–2014q1.

TABLE 1 Vacancy characteristics: AMS universe versus matched sub-sample		
Vacancy characteristics and selected industries	AMS sample	
	Universe	Matched
At least apprenticeship (%)	50.2	46.7
Permanent contract (%)	78.3	80.9
Hired through system (%)	23.3	100.0
Fixed working time (%)	21.3	27.3
Full time (%)	75.5	100.0
Small firm (%)	44.5	40.4
Vienna (%)	17.0	10.2
Manufacturing, mining, and quarrying (%)	10.3	17.1
Construction (%)	7.1	10.6
Wholesale and retail (%)	14.8	14.1
Tourism, hotels, and restaurants (%)	23.4	18.5
Other services (%)	32.2	29.8
Number of vacancies	5,291,897	802,966
Number of establishments	267,122	119,030

Notes: Authors' tabulations of vacancy characteristics using all vacancy outflows for the years 1997–2014 for the AMS vacancy universe and the matched sub-sample. A small establishment is defined as an establishment with 1–employees.

(a) Figure 1: AMS vacancy data versus representative vacancy survey

(b) Table 1: Vacancy characteristics

5.2 Figure 2: Vacancy filling and posting relative to date of availability of job

Read: Panel (a) shows that the filling hazard peaks at the job availability date. Panel (b) shows that many vacancies are posted before the job becomes available.

Economic message. Early posting matters for interpreting vacancy duration and motivates early-posting controls.

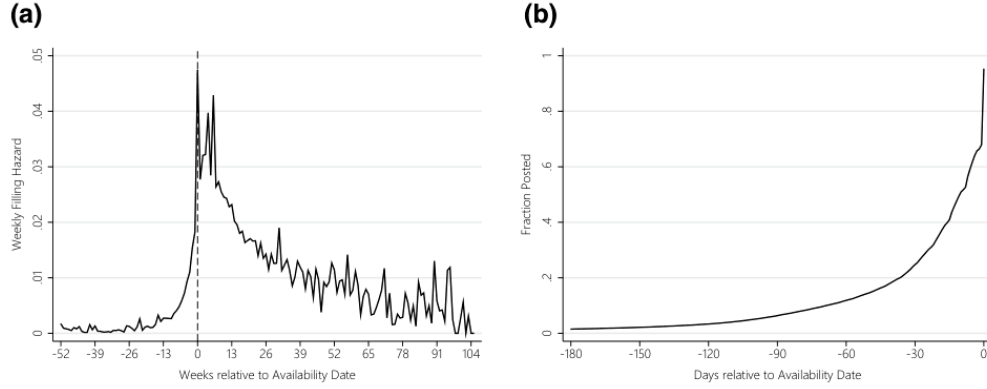


FIGURE 2

Vacancy filling and posting, relative to date of availability of job. (a) Vacancy-filling hazard. (b) Cumulative fraction posted

Notes: Vacancy-filling hazard and cumulative fraction posted relative to date of availability of job in the AMS universe based on exact dates for the years 2007–14.

Figure 2: Vacancy filling and posting relative to date of availability of job

5.3 Figure 3: Log entry wage and log vacancy duration, Table 2: Regressions with log vacancy duration as dependent variable, and Table 3: Worker fixed-effects regressions

Read: Figure 3 shows a positive raw relation. Table 2 shows the relation turns negative after controls; the preferred coefficient is -0.075 . Table 3 shows the result survives worker fixed effects.

Economic message. Raw correlations are misleading because worker and job composition confound the wage–duration relation.

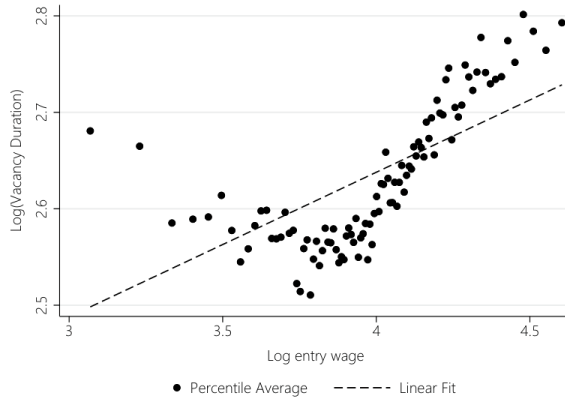


FIGURE 3

Log entry wage and log vacancy duration

Each data point corresponds to the average within a given wage percentile of the sample. The sample is trimmed for a first and shows the 90th percentile.

	Log vacancy duration			
	(1)	(2)	(3)	(4)
Log entry wage	-0.044 (0.015)***	-0.073 (0.016)***		
Log entry wage–log wage floor			-0.123 (0.014)***	-0.162 (0.016)*
Observations	294208	285557	286708	278269
R^2	0.589	0.597	0.593	0.602
Baseline controls	Yes	Yes	Yes	Yes
Additional job and firm controls	No	Yes	No	Yes
Occ. FE (six digits)	No	No	No	No
Individual FE	Yes	Yes	Yes	Yes

Notes: Authors' regressions with the worker sample for the years 1997–2014. Standard errors are clustered at the establishment level. All regressions include a quarterly time fixed effect. Baseline controls include gender, age, age square dummies for the minimum educational requirement of the job and the year of labour market entry, as well as region dummies and an early posting fixed effect, where early posting fixed effects are dummies for posting the vacancy 0 days, 8–30 days, 31–60 days, and more than 60 days prior to the desired start date. Additional job and firm controls include log job duration, on-job wage growth, average worker wage growth, establishment employment growth, log establishment size, and establishment age, see Columns (3) and (5) of Table 2. In Columns (3) and (4), we use the predicted occupation-specific daily wage floors for the full sample period from 1997 to 2014. The wage floors are predicted at the occupation level from the observed wage floors in the AMS vacancy update for the period from 2011 to 2014, see Supplementary Appendix A.1 for more details. The sample is slightly smaller as in Columns (1) and (2) as we do not observe wage floors for all occupations. Levels of significance: *10%, **5%, and ***1%.

(a) Figure 3: Raw bin scatter

(b) Table 3: Worker fixed-effects regressions

TABLE 2
Regressions with log vacancy duration as dependent variable

	Log vacancy duration				
	(1)	(2)	(3)	(4)	(5)
Log entry wage	0.230 (0.014)***	−0.037 (0.009)***	−0.075 (0.010)***		
Log entry wage−log wage floor				−0.065 (0.010)***	−0.104 (0.010)*
Log job duration			0.025 (0.001)***		0.034 (0.002)*
On-job wage growth			−0.067 (0.017)***		−0.052 (0.018)*
Average worker wage growth			0.008 (0.009)		0.007 (0.010)
Establishment empl. growth			−0.012 (0.006)**		−0.018 (0.006)*
Log establishment size			0.029 (0.003)***		0.023 (0.003)*
Establishment age			−0.001 (0.000)***		−0.001 (0.000)*
Observations	535357	535357	522232	520473	507684
R^2	0.010	0.211	0.214	0.172	0.175
Baseline controls	No	Yes	Yes	Yes	Yes
Occ. FE (six digits)	No	Yes	Yes	No	No
Individual FE	No	No	No	No	No

Notes: Authors' regressions with the worker sample for the years 1997–2014. Standard errors are clustered at the establishment level. All regressions include a quarterly time fixed effect (FE). Baseline controls include gender, age, age squared, dummies for the minimum educational requirement of the job and the year of labour market entry, as well as region & industry and an early posting fixed effect, where early posting fixed effects are dummies for posting the vacancy 0–7 days, 8–30 days, 31–60 days, and more than 60 days prior to the desired start date. On-job wage growth is measured as average daily wage growth in the establishment, establishment empl. growth is measured as the monthly change in employment in the vacancy filling month, average worker wage growth is measured as the average daily wage growth of the worker in the next 5 years independent of the establishment. In Columns (4) and (5), we use the predicted occupation-specific daily wage floors for the full sample period from 1997 to 2014. The wage floors are predicted at the occupation level from the observed wage floors in the AMS vacancy update for the period from 2011 to 2014, see Supplementary Appendix A.1 for more details. The sample is slightly smaller as in Columns (1) and (3) as we do not observe wage floors for all occupations. Levels of significance: *10%, **5%, and ***1%.

Table 2: Baseline regressions

5.4 Figures 4–6 and Table 4: AKM wage components and vacancy duration

Read: Figure 4 shows worker components are positively related to vacancy duration. Figure 5 shows establishment wage effects are negatively related to vacancy duration. Figure 6 shows residual wage variation. Table 4 formalizes these patterns; the preferred AKM establishment coefficient is about -0.211 .

Economic message. High-wage workers and high-wage firms have opposite duration patterns. This is the core decomposition result.

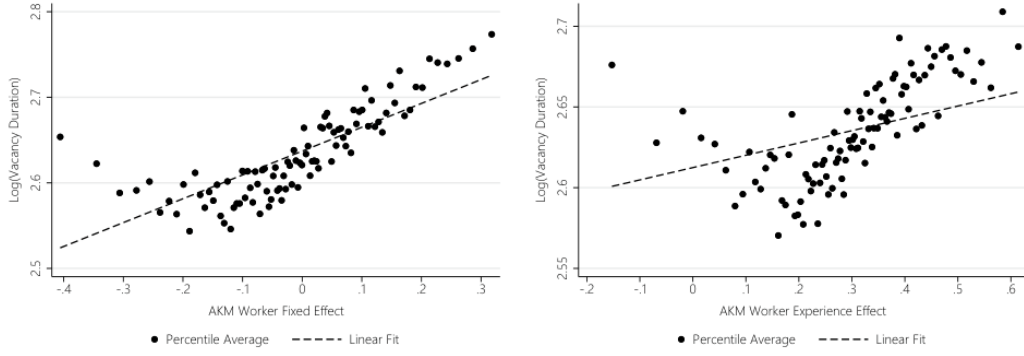


FIGURE 4

AKM worker effects and log vacancy duration

Notes: The figures show partial correlations of AKM worker effects and log vacancy duration, *i.e.* controlling for the AKM establishment effects, the AKM residuals and the AKM person experience effect (left) and the AKM person experience effect (right), respectively. Each data point corresponds to the average within a given percentile of the AKM worker effect in the sample. The sample is trimmed for wage values below the first and above the 99th percentile.

vacation fixed effects and educational requirements, which are unique to the vacancy data and

Figure 4: AKM worker effects and duration

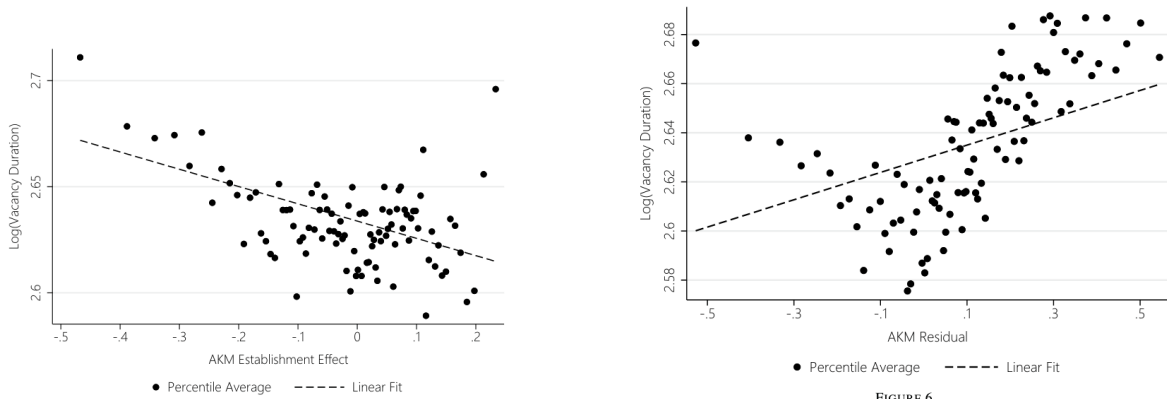


FIGURE 5

AKM establishment effect and log vacancy duration

This figure shows partial correlations of AKM establishment effects and log vacancy duration, *i.e.* controlling for AKM worker and the AKM residual. Each data point corresponds to the average within a given percentile of the AKM establishment effect. The sample is trimmed for wage values below the first and above the 99th percentile.

FIGURE 6

AKM residual and log vacancy duration

This figure shows partial correlations of AKM residuals and log vacancy duration, *i.e.* controlling for AKM worker and establishment effects. Each data point corresponds to the average within a given percentile of the AKM residual in the sample. The sample is trimmed for wage values below the first and above the 99th percentile.

(a) Figure 5: AKM establishment effect

(b) Figure 6: AKM residual

TABLE 4
Regressions with log AMS vacancy duration as dependent variable using AKM effects

	Log vacancy suration					
	OLS				Fixed effects	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. AKM effects						
Log entry wage	0.066 (0.010)***					
AKM establishment effect		−0.082 (0.023)***	−0.098 (0.019)***	−0.211 (0.019)***	−0.158 (0.029)***	−0.269 (0.029)**
AKM worker fixed effect		0.279 (0.013)***	0.057 (0.013)***	0.034 (0.013)**	—	—
AKM worker exp. effect		0.079 (0.015)***	−0.020 (0.011)*	−0.041 (0.011)***	−0.007 (0.029)	−0.011 (0.030)
AKM residual		0.056 (0.010)***	−0.029 (0.009)***	−0.047 (0.010)***	−0.011 (0.016)	−0.017 (0.017)
Observations	486,997	486,997	486,997	475,623	275,967	268,196
R ²	0.178	0.179	0.216	0.219	0.589	0.598
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	Yes	No	Yes
Occ. FE (six digits)	No	No	Yes	Yes	No	No
Individual FE	No	No	No	No	Yes	Yes
Panel B. Avg. establishment earnings						
Log entry wage	0.053 (0.010)***					
Avg. establish. log earnings		−0.007 (0.015)	−0.071 (0.014)***	−0.153 (0.014)***	−0.103 (0.021)***	−0.172 (0.021)*
Residual entry wage		0.090 (0.010)***	−0.021 (0.009)**	−0.037 (0.010)***	−0.015 (0.015)	−0.022 (0.017)
Observations	535,087	535,087	535,087	522,228	294,043	285,555
R ²	0.174	0.174	0.211	0.215	0.589	0.598
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	Yes	No	Yes
Occ. FE (six digits)	No	No	Yes	Yes	No	No
Individual FE	No	No	No	No	Yes	Yes

Notes: Authors' regressions with the worker sample for the years 1997–2014. Standard errors are clustered at the establishment level. All regressions include a quarterly time fixed effect. Baseline controls include gender, age, age squared, dummies for the minimum educational requirement of the job and the year of labour market entry, as well as region, industry and an early posting fixed effect, where early posting fixed effects are dummies for posting the vacancy 0–days, 8–30 days, 31–60 days, and more than 60 days prior to the desired start date. Additional controls include log job duration, on-job wage growth, average worker wage growth, establishment employment growth, log establishment size.

Table 4: AKM effect regressions

5.5 Figure 7: Decision lags versus competition for high-skill workers

Read: Higher-AKM workers are associated with longer vacancy durations but shorter unemployment durations.

Economic message. High-skill jobs may involve thinner applicant pools and longer firm-side decision lags even when workers find jobs quickly.

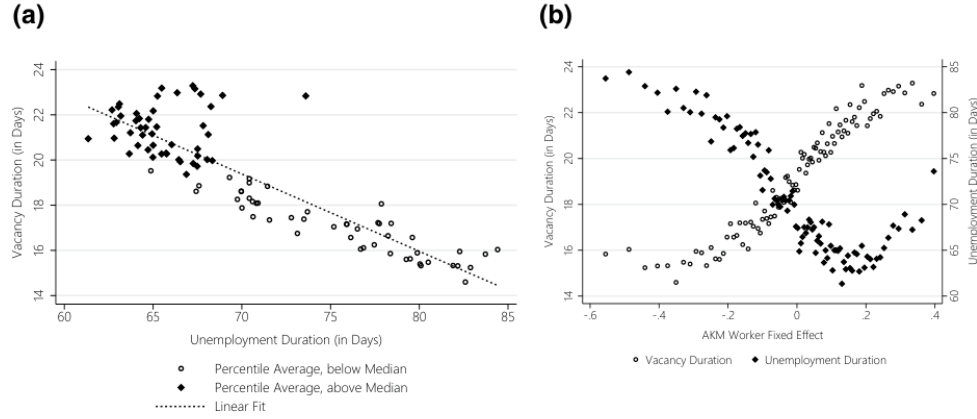


FIGURE 7

Decision lags versus competition for high-skill workers. (a) Vacancy vs. unemployment duration, by AKM worker effect percentiles. (b) Vacancy and unemployment duration, by AKM worker effect percentiles

Notes: The figure shows average vacancy and unemployment duration in days by AKM worker effect percentile in the regression sample of Table 4.

Figure 7: Decision lags versus competition for high-skill workers

5.6 Figure 8: Hires-weighted OLS regression of daily filling hazard and starting wages at the establishment growth bin level and Table 5: Simulation results of Kaas and Kircher (2015)

Read: Figure 8 shows that filling hazards rise with hires rates but wages rise only weakly. Table 5 shows that the directed-search model predicts much larger wage-duration elasticities than the data.

Economic message. Wages cannot quantitatively explain the large relationship between establishment growth and vacancy filling.

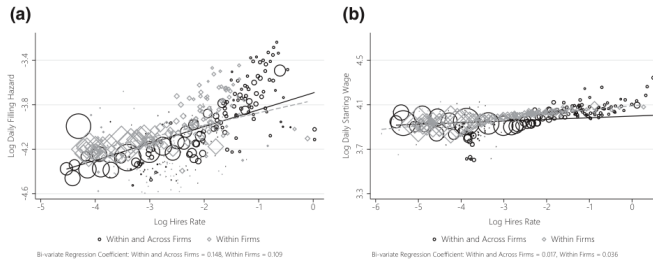


FIGURE 8

Hires-weighted OLS regression of daily filling hazard and starting wages at the establishment growth bin level. (a) Vacancy-filling hazard. (b) Log entry wage

Notes: The figure shows hires-weighted OLS regressions of log daily filling hazards and log daily starting wages against the monthly log hires rate at the establishment growth bin level. We sort establishment-month level observations into bins based on monthly establishment-level DFH employment growth. For each bin, we calculate employment-weighted averages of the log hires rate, days-at-risk weighted averages of the log daily filling hazard and hires-weighted averages of log daily starting wages. We ignore recalls and month where the firm identifier was assigned due to administrative firm identifier changes. Black circles show the weighted averages only controlling for year fixed effects, while grey diamonds additionally control for firm fixed effects. The filling hazard is based on the matched establishment sample, while the starting wage firms is based on the full ACED universe.

(a) Figure 8: Hires-weighted OLS

TABLE 5
Simulation results of model of Kaas and Kircher (2015)

	Data	Model		
		$\gamma = 1$	$\gamma = 0.5$	$\gamma = 0.1$
Elasticity of vacancy duration to starting wage	-0.07	-21.8	-17.9	-3.9
Elasticity of vacancy-filling rate to hires rate	0.15	0.27	0.15	0.03

Notes: This table shows results for the model of Kaas and Kircher calibrated to Austrian data. The statistics from the data refer to the coefficients reported in Column (3) of Table 2 (elasticity of vacancy duration to starting wage) and Figure 8(a) (elasticity of vacancy-filling rate to hires rate). The statistics from the model are based on simulations with 20,000 firms and 400,000 workers over a period of 10 years.

(b) Table 5: Simulation results

6 Short summary

- Raw data show a positive relation between entry wages and vacancy duration.

- After worker and occupation controls, the relation turns negative.
- The preferred entry-wage elasticity is -0.075 ; the preferred AKM establishment-effect elasticity is about -0.211 .
- Worker effects are positively related to vacancy duration, while establishment wage effects are negatively related to duration.
- Wages explain only a small share of filling-rate variation across growing establishments.

7 Conclusion

7.1 Core conclusion

Vacancies posted by higher-wage firms fill faster, but the magnitude of this relationship is small. Wages are a real but quantitatively limited recruiting device.

7.2 What the paper changes conceptually

The paper separates high-wage workers from high-wage establishments. This separation changes the interpretation of wage–duration correlations and shows why raw correlations can be misleading.

7.3 Implication for search-and-matching models

Directed-search and wage-posting models match the sign of the establishment-wage result, but standard quantitative versions predict much stronger wage effects than observed.

7.4 Interpretation of the small elasticity

Recruiting is multidimensional. Search intensity, screening, hiring standards, networks, job amenities, and HR speed likely explain much more of the variation in vacancy filling.

7.5 Policy and measurement implications

Vacancy statistics should be linked to firms and workers whenever possible. Without linked data, researchers may confuse worker selection with firm recruiting effort.

7.6 Limitations and open questions

The paper is not a randomized wage experiment and cannot fully observe non-wage amenities, applicant flows, or recruiting effort. Future work can combine vacancy, application, interview, and wage-posting data.

7.7 Takeaway

The main lesson is that vacancy-filling heterogeneity is not simply a wage-setting problem. Wage policies matter, but the dominant margins of recruiting likely lie elsewhere.